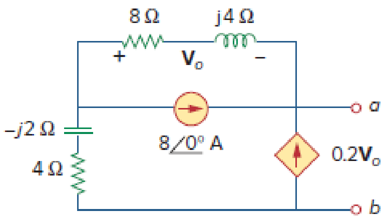
Problem

Determine the Thevenin equivalent of the circuit in Fig. 10.27 as seen from the terminals *a*-*b*.

Figure 10.27:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.27 in the textbook.

Replace the current source by the open circuit to calculate the Thevenin's impedance, $\mathbf{Z}_{\text{Th}}$.

Place a test current source to calculate the Thevenin's impedance, $\mathbf{Z}_{\text{Th}}$.

Calculate the Thevenin's impedance in Figure 1.

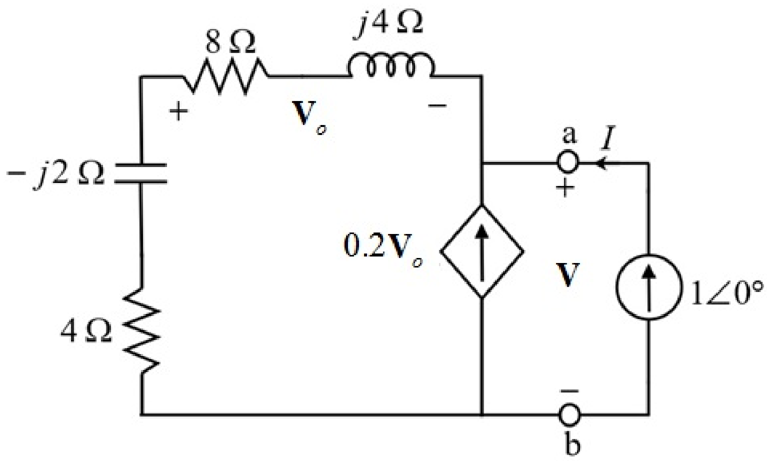


Figure 1

Step 2 of 7

Apply the voltage division and calculate the voltage, $\mathbf{V}_o$.

$$\mathbf{V}_o = -\frac{\mathbf{V}(8+j4)}{(8+j4+4-j2)}$$
$$= -(0.702 + j0.216)\mathbf{V}$$

Apply the nodal analysis at node a in Figure 1.

$$0.2\mathbf{V}_o + 1 = \frac{\mathbf{V}}{(8+j4+4-j2)}$$

Substitute $\mathbf{V}(0.735\angle -162.89°)$ for $\mathbf{V}_o$.

$$-0.2(0.702 + j0.216)\mathbf{V} + 1 = \frac{\mathbf{V}}{(8+j4+4-j2)}$$
$$\mathbf{V}(0.081 - j0.0135 + 0.1405 + j0.0432) = 1$$

$$\mathbf{V} = \frac{1}{(0.2216 + j0.0296)}$$
$$= 4.43 - j0.594$$

Step 3 of 7

Calculate the value of the Thevenin's impedance.

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}}{\mathbf{I}}$$
$$= \frac{4.43 - j0.594}{1\angle 0°}\ \Omega$$
$$= 4.47\angle -7.64°\ \Omega$$

Thus, the Thevenin's impedance is $\boxed{4.47\angle -7.64°\ \Omega}$.

Step 4 of 7

Draw the circuit to calculate the Thevenin's voltage.

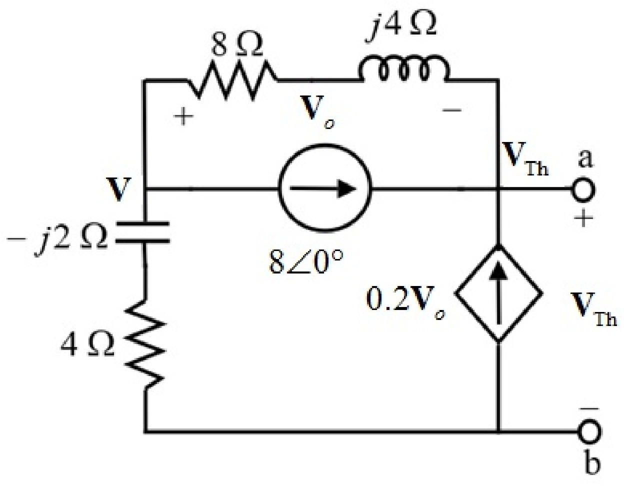


Figure 2

Step 5 of 7

Apply the nodal analysis at node, $\mathbf{V}$.

$$\frac{\mathbf{V}-\mathbf{V}_{\text{Th}}}{8+j4}+8\angle 0°+\frac{\mathbf{V}}{4-j2}=0$$

$$\mathbf{V}\left(\frac{1}{4-j2}+\frac{1}{8+j4}\right)-\mathbf{V}_{\text{Th}}\left(\frac{1}{8+j4}\right)+8=0$$

$$\mathbf{V}(0.3+j0.05)-\mathbf{V}_{\text{Th}}(0.1-j0.05)+8=0$$

$$\mathbf{V}=\left(\frac{0.1-j0.05}{0.3+j0.05}\right)\mathbf{V}_{\text{Th}}-\frac{8}{0.3+j0.05}$$

$$\mathbf{V}=(0.297-j0.216)\mathbf{V}_{\text{Th}}-(25.94-j4.32) \quad \text{......} (1)$$

Step 6 of 7

Apply the nodal analysis at the node, $\mathbf{V}_{\text{Th}}$.

$$\frac{\mathbf{V}_{\text{Th}}-\mathbf{V}}{8+j4}-8-0.2\mathbf{V}_o=0$$

Substitute $\mathbf{V}-\mathbf{V}_{\text{Th}}$ for $\mathbf{V}_o$.

$$\frac{-\mathbf{V}_o}{8+j4}-8-0.2\mathbf{V}_o=0$$

$$\mathbf{V}_o\left(0.2+\frac{1}{8+j4}\right)=-8$$

$$\mathbf{V}_o(0.3-j0.05)=-8$$

$$\mathbf{V}_o=-25.945-j4.324 \text{ V}$$

Step 7 of 7

Consider the expression for $\mathbf{V}_o$.

$$\mathbf{V}_o=\mathbf{V}-\mathbf{V}_{\text{Th}}$$

$$\mathbf{V}=\mathbf{V}_{\text{Th}}+\mathbf{V}_o$$

$$\mathbf{V}=\mathbf{V}_{\text{Th}}-25.945-j4.324$$

Calculate the value of Thevenin’s voltage from equation (1).

$$\mathbf{V}=(0.297-j0.216)\mathbf{V}_{\text{Th}}-(25.94-j4.32)$$

$$\mathbf{V}_{\text{Th}}-25.945-j4.324=(0.297-j0.216)\mathbf{V}_{\text{Th}}-(25.94-j4.32)$$

$$\mathbf{V}_{\text{Th}}(1-0.297+j0.216)=(25.945+j4.324)-(25.94-j4.32)$$

$$\mathbf{V}_{\text{Th}}(1-0.297+j0.216)=5.945\times10^{-3}+j8.644$$

$$\begin{aligned}\mathbf{V}_{\text{Th}}&=\frac{5.945\times10^{-3}+j8.644}{0.703+j0.216}\\&=3.46+j11.23\\&=11.754\angle 72.88° \text{ V}\end{aligned}$$

Thus, the Thevenin’s voltage is $\boxed{11.754\angle 72.88° \text{ V}}$.